the JAR file to *'libs'*. Right-click the *'libs'* folder in Eclipse, and select *'Configure Build Path'*. Add the PhoneGap JAR file from the project. Close the dialogue and refresh the project by hitting F5.

Finally, we will modify two files, *'AndroidManifest.xml'* and *'App.java'*, before our project is ready to run. In Eclipse, right-click *'AndroidManifest.xml'* and choose *'Open With > Text Editor'*. Look for the following snippet at the beginning of the file:

```
<manifest xmlns:android="http://schemas.android.com/apk/res/
android"
      package="com.phonegap.helloworld"
      android.versionCode="1"
      android.versionName="1.0">
```

Paste the following below the located snippet (copy from the *manifest-additions.txt* file on the *LFY* website at http://linuxforu.com/article_source_code/aug11/phonegap.zip).

```
<supports-screens
      android:largeScreens="true"
      android:normalScreens="true"
      android:smallScreens="true"
      android:resizeable="true"
      android:anyDensity="true"
/>
<uses-permission android:name="android.permission.CAMERA" />
<uses-permission android:name="android.permission.VIBRATE" />
<uses-permission android:name="android.permission.ACCESS_COARSE_
LOCATION" />
<uses-permission android:name="android.permission.ACCESS_FINE_
LOCATION" />
<uses-permission android:name="android.permission.ACCESS_
LOCATION_EXTRA_COMMANDS" />
<uses-permission android:name="android.permission.READ_PHONE_
STATE" />
<uses-permission android:name="android.permission.INTERNET" />
<uses-permission android:name="android.permission.RECEIVE_SMS" />
<uses-permission android:name="android.permission.RECORD_AUDIO"
/>
<uses-permission android:name="android.permission.MODIFY_AUDIO_
SETTINGS" />
<uses-permission android:name="android.permission.READ_CONTACTS"
/>
<uses-permission android:name="android.permission.WRITE_CONTACTS"
/>
<uses-permission android:name="android.permission.WRITE_EXTERNAL_
STORAGE" />
<uses-permission android:name="android.permission.ACCESS_NETWORK_
STATE" />
```

To the *activity* tag in the XML file, append *android:configChanges="orientation|keyboardHidden"*, so it now looks like the following code snippet:

```
<activity android:name=".helloworld"
      android:label="@string/app_name" android:configChanges="orien
tation|keyboardHidden" >
```

Make sure you keep saving the project, to avoid any data loss. Now, open the *'src/com.phonegap.helloworld/App.java'* file in Eclipse, and to the import lines at the top, add:

```
import com.phonegap.*;
```

Next, modify the class *App* to extend *'DroidGap'*. The line would then look like what's shown below:

```
public class App extends DroidGap {
```

Last, reference your *index.html* file. Here, you have replaced the *setContentView()* method with the following line below the *onCreate()* method:

```
super.loadUrl("file:///android_asset/www/index.html");
```

That completes the process. Now run the app in the emulator, using the AVD you had created earlier. Right-click the project, and select *'Run As > Android Application'*. Eclipse will ask you to select the AVD, and will then load and run the project. Note that the AVD will take a while to load, especially the first time you start it.

You have successfully created your first Android app, with only a little HTML and some fiddling with Java code in Eclipse!

Exploring beyond this can get very interesting, let me warn you. Jonathan Stark has written a nice little book on making Android apps using HTML5/CSS/Javascript. You can read the book for free (link given below). Also, to make Web applications look like native apps, do head over to jQtouch's website. This is a nifty little JS library that makes your HTML5 app look slick and cool, just as if you made it using native code. What's more, you can also exploit the native hardware capabilities using the same techniques. Cool, right? So, what are you waiting for? END

References

1. Building Android Apps with HTML, CSS, and JavaScript by Jonathan Stark: *http://ofps.oreilly.com/titles/9780596805784/*
2. PhoneGap: *http://www.phonegap.com*
3. jQTouch: *http://jqtouch.com*
4. Google Android Developer Home: *http://developer.android.com*
5. Eclipse IDE Home: *http://www.eclipse.org*
6. OpenJDK: *http://openjdk.java.net/*

By: Siddharth Mankad

The author is a new media designer, with an affinity for physical computing, computer software and code, and country music.